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Effects of exposure on perceptual evaluations of dialect-specific allophonic variation

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ABSTRACT:

Dialects vary in their allophonic patterns, potentially affecting listeners' cognitive representations of language. How different exposure (in terms of size of acoustic distinction) to dialect-specific allophonic patterns for two American English vowels, /æ ai/, affects listeners' behaviors in a perceptual similarity judgments task is explored. Theories about the phonologization of allophonic patterns, which have largely relied on production data, using novel perception data are examined. Listeners from northeastern Ohio (North), a region with less /æ/ allophony and more /ai/ allophony, and listeners from central and southwestern Ohio (Midland), a region with more /æ/ allophony and less /ai/ allophony, are contrasted. Contextual allophones of the same phoneme sound more similar to listeners with less exposure to dialect-specific allophonic variation than they do to listeners with more exposure. Exposure influences the degree of phonological contrast present in listeners' cognitive representations of allophones. These results support late abstractness theories of phonologization, in which phonological abstractness gradually develops during a sound change in progress in a community. © 2026 Acoustical Society of America. <https://doi.org/10.1121/10.0042426>

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I. INTRODUCTION

Many vowels in American English dialects show substantial allophonic variation in production, in which acoustically distinct but not phonologically contrastive variants appear in specific phonological conditioning environments (Labov *et al.*, 1991; Labov, 1994). Listeners' exposure to this variation has the capacity to shape their cognitive representations of sound categories, as existing evidence suggests that these representations are sharply calibrated to the specific acoustic details of listeners' dialect exposure (Clopper, 2014; Clopper and Walker, 2017; Ranbom and Connine, 2007; Sumner and Samuel, 2009). In turn, these cognitive representations of allophones impact how listeners interpret and comprehend the incoming auditory signal when perceiving speech. For example, listeners can leverage these representations to facilitate spoken word recognition (Farris-Trimble and Tessier, 2019; Mitterer *et al.*, 2018;), even when allophones occur in a second language (Dailey *et al.*, 2024) or in a second dialect (Dahan *et al.*, 2008; German *et al.*, 2013). While many studies have focused on what allophones can do for listeners during spoken word recognition, in this paper, I explore the role that dialect exposure plays in shaping the development of cognitive representations of allophonic variation and examine how these representations are accessed and applied during a perceptual similarity experiment. I show that higher dialect exposure (i.e., exposure to larger acoustic differences across allophones) facilitates the development of more phoneme-like representations of allophonic categories, with these representational

differences warping listeners' perceptions of acoustically identical stimuli.

A. Allophonic variation and phonological contrast

Listeners perceive different kinds of sounds as having variable amounts of perceptual similarity to one another according to their language's specific phonological structures, an observation first noted by Trubetzkoy (1939). For example, Japanese listeners with minimal English experience have difficulty distinguishing the English phonemes /l ɹ/, presumably because both sounds belong to a single phoneme category in the Japanese language and are therefore highly perceptually similar (MacKain *et al.*, 1981). Similar effects have been shown for other phoneme and language pairs, such as English listeners' difficulty distinguishing Hindi dental and retroflex stops (Werker *et al.*, 1981) or Cantonese listeners' difficulty distinguishing Russian palatalized liquids (Smith and Kochetov, 2019). Taken together, these results suggest that listeners are highly perceptually attuned to the phonological categories employed by their native languages but face some difficulty when prompted to make perceptual distinctions within these categories.

However, languages also contain phonological categories that are not strictly phonemic distinctions, like allophones [see Hall (2009) for an overview of intermediate phonological categories]. Boomershteyn *et al.* (2008) studied how listeners perceived allophonic distinctions versus phonemic distinctions. They reported that American English listeners perceived sounds in different phoneme categories (e.g., [t d]) as being less similar than sounds in the same phoneme category (e.g., allophones, such as [t r]). Sounds

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in the same phoneme category (i.e., allophones) were also perceived as being less similar than an identity case (e.g., [t t]), indicating that listeners attend to allophonic variation and that allophones fall somewhere between maximal phonological contrast (phonemes) and minimal phonological contrast (identity). While the evidence of [Boomershine et al. \(2008\)](#) suggests that allophonic variation does not carry the same representational status as phonemic variation, there is a lot of representational space available for allophones to exist in on the continuum between maximal and minimal phonological contrast.

The mechanism by which allophones gain this special intermediate status in perception is an open question. In the realm of phonemic distinctions, [MacKain et al. \(1981\)](#) critically found that additional exposure to English improved Japanese listeners' discrimination of /l ɹ/, suggesting that exposure can facilitate the construction of new (contrasting) representations of speech sounds. That is, listeners can gain perceptual sensitivity within the acoustic space of a single phoneme category in their native language so long as they have adequate exposure to what the new phonemic categories are supposed to be in the second language. This sort of development is straightforwardly explained in exemplar models of linguistic representations (e.g., [Goldinger, 1998](#)), in which exposure allows new phonological categories to emerge organically based on the shared acoustic characteristics of individual episodic traces. I argue in this paper that this exposure mechanism for new phonemic category development could also reasonably be applied to the acquisition of allophonic (i.e., sub-phonemic) categories, such that exposure to adequate acoustic divergence within a single phonemic category spurs listeners to develop two distinct representational categories for two allophones of a single phoneme.

An alternative viewpoint on representations of allophonic variation is that exposure to more divergent acoustic productions of allophones should prompt less, not more, attention paid to allophonic variation. This viewpoint relies on the notion that these listeners do not need to pay a lot of attention to these acoustic details when listening to their own dialect. However, listeners who have exposure to less divergent acoustic productions of allophones may have been trained to pay attention to this smaller difference. This possibility is difficult to resolve with existing evidence that exposure promotes new phonological category development [e.g., the findings of [MacKain et al. \(1981\)](#) that exposure to English improves Japanese listeners' discrimination of /l ɹ/].

Some researchers have proposed that certain cases of allophonic variation can become phonologized or even phonologically contrastive ([Hualde et al., 2017](#); [Mielke et al., 2003](#); [Moreton and Thomas, 2007](#); [Sneller, 2018](#)). One such case is pre-voiceless /ai/ raising, an allophonic split in which the raised allophone [ɰi] appears before voiceless obstruents and the unraised allophone [ai] appears elsewhere. This case of allophony has been described as phonological because most individuals with this allophonic split produce the raised allophone [ɰi] according to the following sound's

underlying voicing rather than its surface voicing, indicating that the pattern is phonologically abstract. This phonologically opaque /ai/ raising pattern is characterized by raised allophone [ɰi] in both *write* and *writing*. Because of its interaction with /t/ flapping in English, this type of /ai/ raising dialect generates minimal pairs like [ɰaɪrɪŋ] for *writing* and [ɰaɪrɪŋ] for *riding* that are only distinguished by [ɰi] versus [ai]. A minimal pair-based account of phonological contrast maintains that the raised allophone [ɰi] constitutes a marginal phoneme in this scenario ([Hualde et al., 2017](#)).

Another case of allophonic variation that has been described as phonologized or even phonologically contrastive is the nasal split /æ/ system, in which pre-nasal [æ̃] is much higher and fronter than pre-oral [æ] ([Boberg and Strassel, 2000](#); [Kiparsky, 1995](#); [Labov et al., 2016](#)). This allophonic pattern can interact with word-final unreleased stops to generate minimal pairs like [bæ̃ʔ] for *ban* and [bæʔ] for *bad*. Additional evidence suggests that listeners perceive [bæ̃d] as “band” even though the [n] has been fully removed, demonstrating that vowel nasalization can be perceived as contrastive by English listeners in the absence of consonantal cues ([Zellou, 2017](#)). Again, a minimal pair-based account of phonological contrast would conclude that [æ̃] is a marginal phoneme based on this distributional evidence.

Such a phoneme would be marginal in terms of its functional load (that is, it is involved in relatively few lexical contrasts), but it is reasonable to expect that listeners' phonological representations would be sensitive to whether an allophone has phonologized in their language. Listeners with a phonologized system would be expected to treat matching allophones (same vowel, same allophone) and mismatching allophones (same vowel, different allophone) as less similar as compared to listeners without a phonologized system. In other words, mismatching allophones would behave more like mismatching phonemes for listeners with a phonologized system than for listeners without a phonologized system because those allophones have a higher amount of phonological contrast due to phonologization.

However, it is unclear whether the existence of minimal pairs in production data is an appropriate diagnostic for phonologization of allophonic variation [see [Halle \(1962\)](#) for a discussion of taxonomic versus systematic phonemes]. In addition to this theoretical controversy around minimal pairs, relatively rare instances of marginal phonemes may not strongly signal anything about listeners' broader representational systems. The phonological status of allophonic variation is unlikely to be fully accessible in production data alone, given that phonologization has been defined as a cognitive process entailing changes to phonological representations ([Beddor, 2009](#); [Hyman, 1976](#)) and that the nature of these representations has historically been investigated with speech perception data ([Connine et al., 2008](#); [Dahan et al., 2008](#); [Goldinger, 1998](#); [Ranbom and Connine, 2007](#); [Sumner and Samuel, 2009](#)). Perception data provides an opportunity to explore listeners' cognitive representations of allophonic variation from a more systemic angle, bypassing

the controversy of how to interpret the phonological significance of minimal pairs in production data altogether.

One methodology that has been employed to investigate perceptions of allophonic variation is a minimal pair reading task (Herold, 1997) that is aimed at eliciting perceptual similarity judgments. Nesbitt (2023) employed a variant of this type of task by prompting participants to read pairs of words aloud to themselves and evaluate how similar the vowels in each word pair sounded to them (“same” or “different”). The stimuli included several types of word pairs, including match (*hand-ham*), allophonic mismatch (*hand-half*), and phonemic mismatch (*hand-hoop*). She compared this perceptual similarity data to production data from the same participants. Her results showed that participants who produced an acoustically stronger nasal split /æ/ system also perceived mismatch pairs as less similar than participants who produced an acoustically weaker nasal split /æ/ system. She observed that listeners at different stages of an ongoing phonological change in their community showed different perceptions of allophonic variation. Therefore, she concluded that dialect exposure to a phonological change has empirical consequences for listeners’ perceptual systems because it can cause participants to develop variable cognitive representations of allophonic variation.

One alternative interpretation of Nesbitt’s (2023) findings is that participants faithfully or veridically perceived the acoustic characteristics of their own productions of pre-nasal versus pre-oral /æ/ allophones. That is, since participants were evaluating their own productions, it makes sense that participants with an acoustically stronger nasal split /æ/ system would perceive their /æ/ allophones as less similar than participants with an acoustically weaker nasal split /æ/ system. Psychoacoustic research has established that American English listeners can detect a just-noticeable difference in /æ/ allophones that are around 30 Hz apart in F1 (Kewley-Port and Watson, 1994), and the range of /æ/ tokens in Nesbitt’s (2023) study far exceeded this just-noticeable difference threshold. Therefore, it is worth asking whether the participants in her study were just faithfully or veridically reporting the actual acoustic differences in their own speech. For example, a participant perceiving a 200 Hz difference in F1 across /æ/ allophones should find these tokens to be more different than if they were perceiving a 50 Hz difference in F1 across /æ/ allophones for straightforward psychoacoustic reasons. While this perceptual difference could be attributable to different cognitive representations of allophonic variation, as Nesbitt (2023) proposed, it may also be the result of faithful or veridical perception regardless of whether representational differences are present across listeners with different dialect exposure to a phonological change.

In this study, I modified Nesbitt’s (2023) task such that listeners with different dialect exposure to allophonic variation evaluated identical auditory stimuli, controlling for the confound of faithful or veridical perception that is present in her study. This modification allowed for a more direct examination of how listeners’ cognitive representations

impact their perceptions of allophonic variation by bypassing variation in production altogether, isolating effects of representational categories alone. If listeners with different dialect exposure to allophonic variation harbor cognitive representations of allophonic variation that are more or less phonologically contrastive and therefore categorical, then differences in perceptual behaviors (including perceptual similarity judgments) should be observable. These differences should be observable even and especially with controlled auditory stimuli because listeners’ existing representations are expected to exert a top-down influence on or warp their perceptual processing and interpretations of those stimuli. The goal of this study is to investigate representational differences across listeners with different dialect exposure to allophonic variation, focusing on revealing effects of these differences on perceptions of allophonic variation.

II. APPROACH

To understand how dialect exposure affects perceptions of allophonic variation, I compare the perceptual behaviors of listeners with different dialect backgrounds for allophony in two linguistic variables: /ai æ/. American English listeners who are lifelong residents of the Midland and Northern regions of Ohio provide a neat test case for this comparison. The locations of these two dialect regions within Ohio are shown in Fig. 1. The Midland dialect region, encompassing roughly the Cincinnati, Columbus, and Dayton metropolitan areas (U.S. Census Bureau, 2020), is shown in lighter shading. The Northern dialect region, encompassing roughly the Akron, Canton-Massillon, Cleveland, and Youngstown-Boardman metropolitan areas (U.S. Census Bureau, 2020), is shown in darker shading. These boundaries were selected roughly following Labov *et al.* (2006).

The two linguistic variables of interest, /ai æ/, have inverse dialect-specific patterns of allophonic variation across these two regions. Therefore, it is possible to fully cross-exposure to allophonic variation for a linguistic variable with dialect region, such that each listener group has more exposure to allophonic variation for one linguistic variable but less exposure to allophonic variation for the other linguistic variable. This design choice centers dialect exposure as a unifying explanatory mechanism for variability in perceptual behaviors.

A. /ai/ variable

The English diphthong /ai/ is well-known for being involved in an allophonic split in which the raised allophone [ai] appears before voiceless obstruents and the unraised allophone [ai] appears elsewhere. Joos (1942) first observed /ai/ raising among Canadian schoolchildren, noting that it could have emerged from effects of pre-fortis clipping on vowel durations in English (see also Bermúdez-Otero, 2003; Pycha and Dahan, 2016). Since the study of Joos (1942), this feature has been observed in a wide variety of Canadian and American English dialects, earning it the labels



FIG. 1. The counties included in the Midland (lighter) and Northern (darker) dialect regions in the state of Ohio.

“Canadian raising” (Chambers, 1973) and “American raising” (Davis and Berkson, 2021), respectively. In Canadian English dialects, this phenomenon is geographically widespread and relatively diachronically old (Chambers, 1973; Labov *et al.*, 2006). For a long time, researchers even speculated that /ai/ raising existed exclusively in Canadian English (Chambers, 1973; Householder, 1983). More recent studies have dispelled this speculation by showing that this feature is present in many American English dialects, especially in and near the Northern dialect region (Berkson *et al.*, 2017; Dailey-O’Cain, 1997; Hualde *et al.*, 2017; Moreton and Thomas, 2007; Vance, 1987). The presence or absence of /ai/ raising has been conventionally established as a 60 Hz difference in F1 in pre-voiceless versus pre-voiced environments in monosyllables (Labov *et al.*, 2006).

This study focuses on American English speakers who are lifelong residents of Ohio, who have variable qualities of dialect exposure to /ai/ raising by virtue of living in different dialect regions within the state. I treat acoustic quality of exposure to mean the size of the acoustic difference between pre-voiceless and pre-voiced obstruent /ai/ allophones in a

dialect. This factor varies across the Northern and Midland regions of Ohio, so I treat residents of these two dialect regions as having “higher” or “lower” exposure to /ai/ allophony, respectively.

The Northern dialect region is characterized by widespread /ai/ raising. In my word list recordings of lifelong Northern Ohioans ($M = 21.14$ years old at time of recording in 2023, $n = 33$), they produced an average difference of 139 Hz in the F1 nucleus (25% of vowel duration) of /ai/ allophones across pre-voiceless and pre-voiced obstruent environments. All 33 people recorded exceeded the 60 Hz threshold of Labov *et al.* (2006) for diagnosing /ai/ raising. In contrast, the Midland dialect region is characterized by /ai/ raising that is less acoustically developed. In my word list recordings of younger lifelong Midland Ohioans ($M = 21.14$ years old at time of recording in 2023, $n = 50$), they produced an average difference of 56 Hz in the F1 nucleus (25% of vowel duration) of /ai/ allophones across pre-voiceless and pre-voiced obstruent environments. Only 24 people (48%) met or exceeded the 60 Hz threshold of Labov *et al.* (2006) for diagnosing /ai/ raising. Diachronically newer /ai/ raising systems are characterized

by less acoustic differentiation between the two /ai/ allophones, since older /ai/ raising systems have undergone phonetic enhancement over time [e.g., Fig. 3 in [Fruehwald \(2016\)](#)]. My word list recordings show clear evidence of phonetic enhancement in Northern Ohio, with an average difference in the F1 nucleus of /ai/ allophones of 139 Hz as compared to only 56 Hz in the Midland. Therefore, the acoustic quality of listeners' exposure to /ai/ raising in the North is increased relative to the Midland due to these acoustic differences.

Although it is not the primary focus of this study, it is also worth noting that the distribution of /ai/ raising in terms of number of speakers varies somewhat across these two dialect regions. In the Northern dialect region, /ai/ raising has been documented for people born as early as 1878 ([Moreton and Thomas, 2007](#); [Thomas and Mielke, 2021](#)) and is uniformly present among the younger lifelong Northern Ohioans that I sampled who were born in the late 1990s to early 2000s ($n = 33$). In the Midland dialect region, little /ai/ raising was documented among people born from 1977–1991 ([Thomas, 2001](#)), though it was present in around half of the younger lifelong Midland Ohioans I sampled who were born in the late 1990s to early 2000s ($n = 50$). This difference is due to /ai/ raising being an incipient change in progress in many cities in the Midland dialect region ([Berkson et al., 2017](#)), leading to an age-graded pattern that is characterized by a high level of variability in the younger generation [see [Beddor \(2023\)](#) for a review]. While the current study focuses on differences in the acoustic quality of /ai/ raising across the two regions, the exact nature of listeners' exposure in terms of number of speakers is also variable due to how /ai/ raising diachronically develops.

There is also some inter-speaker variation in the distribution of the raised allophone [ɪ] with respect to its interaction with post-tonic /t d/ flapping in English, potentially affecting its status as phonologized or phonological. [Joos \(1942\)](#) observed two production patterns among Canadian schoolchildren in Ontario, one in which *typewriter* was produced as [tɪpɪɪrɪ] (“Dialect A”) and another in which *typewriter* was produced as [tɪpɪɪrɪ] (“Dialect B”). Whether or not the surface voiced flap [ɾ] blocks the raised allophone [ɪ] from appearing, and what this variability means for linguistic theory, has been a subject of much interest and agony for phonologists (explaining the characterization of it by [Joos \(1942\)](#) as a “phonological dilemma”).

Dialect A has been widely described as being phonologized or phonological, since the raised allophone [ɪ] is strictly conditioned by the underlying voicing of the following sound ([Joos, 1942](#)). That is, the following surface voiced flap [ɾ] does not block the raised allophone [ɪ] because the following environment is underlyingly voiceless. In contrast, Dialect B has been widely described as phonetic, since the raised allophone [ɪ] is strictly conditioned by the surface voicing of the following sound ([Joos, 1942](#)). That is, the following surface voiced flap [ɾ] blocks the raised allophone [ɪ] because the following environment is not surface voiceless, even though it is underlyingly voiceless.

There are two major theories about the phonological development of /ai/ raising, both of which are largely derived from production data. The first theory, conventionally labeled “early abstractness,” holds that /ai/ raising is fully phonological from its outset in a community ([Fruehwald, 2016](#)). [Fruehwald \(2016\)](#) reported that lifelong residents of Philadelphia, Pennsylvania only ever produced Dialect A, a phonologically opaque raising pattern in which the raised allophone [ɪ] appears in both *write* and *writing*. He did not locate any empirical evidence that Dialect B, in which the raised allophone [ɪ] appears in *write* but is blocked by the surface voiced flap [ɾ] in *writer*, ever existed in the community. Therefore, he argued that Dialect A was present from the outset of the phonological change in the community, suggesting that a fully phonologically abstract (i.e., opaque) pattern was innovated immediately. This viewpoint is consistent with “big bang” and substance free theories of phonological change ([Janda, 2003](#); [Anderson, 1981](#)), in which individual phonologies can be rapidly updated nearly instantaneously and must not be directly motivated by phonetic factors (e.g., coarticulation). Further, [Fruehwald \(2016\)](#) asserted that the reason for the rapid development of Dialect A among Philadelphians was that the division between pre-voiceless and pre-voiced /ai/ allophones was already present in individual listeners' phonologies due to representations of pre-fortis clipping in English. Following this line of logic, /ai/ raising has an explicitly phonological origin and, therefore, should always be subject to strict phonological conditioning, even when production differences in a community are only beginning to emerge.

The second theory, conventionally labeled “late abstractness,” characterizes /ai/ raising as gradually accumulating phonological abstractness over time when it is developing a community ([Berkson et al., 2017](#); [Davis et al., 2020](#)). One consequence of this theoretical viewpoint is that listeners' representations of an ongoing phonological change are not static. Rather, listeners' cognitive representations start out as less abstract and eventually become fully phonologically abstract or phonologized. This perspective crucially contrasts with the perspective of [Fruehwald \(2016\)](#) on the development of phonological representations. Specifically, the two theories differ in whether listeners with dialect exposure to an earlier stage of a phonological change should have similar or dissimilar cognitive representations of allophonic variation as compared to listeners with dialect exposure to a later stage of a phonological change.

Comparing perception data from Midland versus Northern listeners in Ohio can distinguish these two theories of phonologization because they make different predictions about effects of exposure on listeners' cognitive representations of allophonic variation. If phonological abstractness is early, then listeners' cognitive representations of allophonic variation should be comparable across the two dialect regions even when listeners have different qualities of exposure to that allophonic variation. If phonological abstractness is late, then listeners' cognitive representations of allophonic variation should differ across the two dialect

regions, since listeners with higher exposure (i.e., exposure during a later stage of phonological change in which the allophones are more acoustically distinct) should have more abstract cognitive representations of allophonic variation than listeners with lower exposure (i.e., exposure during an earlier stage of phonological change in which the allophones are less acoustically distinct). Specifically, Northern listeners, who have higher dialect exposure to allophonic variation for the /ai/ variable, should perceive mismatching allophones as more like mismatching phonemes (or, less like matching allophones) than Midland listeners, who have lower dialect exposure to allophonic variation for the /ai/ variable.

B. /æ/ variable

The English vowel /æ/ shows allophonic variation in pre-nasal versus pre-oral environments in many dialects, and this variation is conventionally described as a nasal split /æ/ system (Nesbitt, 2023; Zellou and Scarborough, 2019). The nasal split /æ/ system is thought to be the most widespread allophonic system for /æ/ throughout American English dialects (Labov *et al.*, 2006), appearing in dialect regions such as the Mid-Atlantic (Becker and Wong, 2010; Labov *et al.*, 2016), the North (Dinkin, 2022; D’Onofrio and Benheim, 2020; Nesbitt, 2023; Wagner *et al.*, 2016), and the Midland (Boberg and Strassel, 2000; Durian, 2012). This system often contrasts with a more localized /æ/ system, like TRAP raising in areas affected by the Northern Cities Shift (Labov *et al.*, 1972). Specifically, TRAP raising involves the raising of /æ/ in pre-oral environments, rendering pre-oral /æ/ more acoustically similar to pre-nasal /æ/ in this system than in the nasal split /æ/ system.

Residents of the Northern region in Ohio largely produce TRAP raising in accordance with the conventions of the Northern Cities Shift (see map 11.8, Labov *et al.*, 2006). Younger residents of other parts of the North, like Chicago, Illinois (D’Onofrio and Benheim, 2020); Lansing, Michigan (Nesbitt *et al.*, 2019; Nesbitt, 2023); and upstate New York (Dinkin, 2022; Thiel and Dinkin, 2020), have rapidly retreated from TRAP raising and adopted the nasal split /æ/ system instead due to social stigmatization of TRAP raising in those areas. This change accompanied a retreat from other conventional features of the Northern Cities Shift, as well. However, in the Northern region of Ohio, there is evidence that people have not retreated from the TRAP raising system in the same manner as elsewhere in the Northern dialect region. Residents of the Northern region in Ohio either do not view their own speech as different from the speech in Ohio’s Midland region or view their own speech as standard relative to the speech in Ohio’s Midland region (Campbell-Kibler, 2012; Campbell-Kibler and Bauer, 2015). Therefore, the impetus for the reversal of TRAP raising is not present in Ohio’s Northern region. Lifelong Northern Ohioans should be expected to have higher exposure to TRAP raising, in which /æ/ allophones are less acoustically distinct than in the nasal split /æ/ system, than lifelong Midland Ohioans.

Although the nasal split /æ/ system and TRAP raising both involve some level of allophonic variation in the /æ/ vowel, the acoustic quality of this variation is not equivalent. The acoustic difference between the two /æ/ allophones is larger in the Midland than in the North due to the presence of TRAP raising in the North. Therefore, I treat listeners with higher exposure to the nasal split /æ/ system (i.e., Midland listeners) as having higher exposure to allophonic variation in the /æ/ vowel than listeners with higher exposure to TRAP raising (i.e., Northern listeners). This exposure pattern for allophonic variation in the /æ/ vowel is the inverse of these listeners’ higher versus lower exposure to /ai/ raising, fully crossing exposure with dialect region in the experimental design.

There are two major theories about the phonological status of the nasal split /æ/ system that parallel the two theories about the phonological development of /ai/ raising, though the nasal split /æ/ system is not proposed to be a change in progress. The first theory maintains that the pre-nasal /æ/ allophone is a phonologized coarticulatory effect, with early abstractness of phonological status in listeners’ cognitive representations. Many researchers have argued in favor of this position (Boberg and Strassel, 2000; Kiparsky, 1995; Labov *et al.*, 2016). Kiparsky (1995) proposed that the nasal split /æ/ system is phonological because the two allophones appear to be generated by some sort of phonological rule. This idea about what it means for something to be phonological requires a certain set of assumptions about the nature of phonological categories. For example, Labov *et al.* (2016) proposed that abstract phonological categories (e.g., “tense”) motivated diachronic changes to the /æ/ system in Philadelphia, Pennsylvania, suggesting that the community’s developing /æ/ system was directly related to and in fact caused by the phonology all along. This line of thinking closely mirrors what Fruehwald (2016) proposed about /ai/ raising, which is that it originated in elements of the phonology itself, and therefore, it has always been phonologized.

The second theory parallels the late abstractness theory of /ai/ raising, in which allophonic variation in the /æ/ vowel becomes more phonologically abstract over time. The raising of the pre-nasal /æ/ allophone could merely be a coarticulatory consequence of the following nasal consonant for some amount of time before gaining a more abstract phonological representation. Several researchers have observed that the presence of a nasal formant does generate a percept of lowered F1 (i.e., raising) in pre-nasal /æ/ for listeners (Beddor *et al.*, 1986; Carignan *et al.*, 2011). However, this theory has received considerably less attention than the first theory, primarily because acoustic discreteness and separation of allophones in production data have been treated as robust indicators of phonologization (Labov *et al.*, 2016; Nesbitt, 2023). It is unclear whether acoustic discreteness and separation in production data are an appropriate diagnostic for the phonological status of a nasal split /æ/ system, though.

C. Predictions

Whether or not allophonic variation is phonological or phonologized can be tested by examining whether listeners with higher versus lower exposure to allophonic variation for a specific linguistic variable vary in how closely their allophonic perception behaviors mirror their phonemic perception behaviors. If allophonic variation were to be phonologized in a dialect, then those listeners should treat allophonic distinctions more like phonemic distinctions than listeners with lifelong exposure to a non-phonologized dialect. In this study, I examined listeners' evaluations of three word pair types for vowel perceptual similarity: match pairs, allophonic mismatch pairs, and phonemic mismatch pairs. Table I shows definitions and representative sample words for each of these three pair types.

First, I predicted that there would be a tripartite distinction in perceptual similarity ratings across the three pair types. This result would be consistent with the findings of Boomershine *et al.* (2008) findings, which were that phonemic mismatches were perceived as less similar than allophonic mismatches and that allophonic mismatches were perceived as less similar than matches. This result is expected, given what is already known about perceptions of amounts of phonological contrast in perceptual similarity tasks.

Second, I predicted that listeners with higher dialect exposure to allophonic variation for a linguistic variable would rate allophonic mismatches as less similar to matches than listeners with lower dialect exposure to allophonic variation for that variable. That is, listeners with higher dialect exposure to allophonic variation for a linguistic variable would treat allophonic mismatches as more similar to phonemic mismatches than listeners with lower dialect exposure to allophonic variation for that variable. Some research suggests that dialect exposure to higher levels of acoustic overlap among allophones makes them less perceptually distinguishable from one another for listeners (Bowie, 2000; Nesbitt, 2023). Other research has shown that allophones are less likely to be explicitly represented in listeners' phonologies without adequate exposure (Dahan *et al.*, 2008; Sumner and Samuel, 2009). Taken together, these results indicate that exposure should promote more categorical

differentiation of mismatching allophones, consistent with late abstractness theories of the phonologization of allophonic variation.

One alternative possibility is that there could be no difference in perceptual behaviors according to dialect exposure. This finding would be consistent with early abstractness theories, in which higher versus lower exposure is irrelevant to the nature of the phonological representation due to the early development of phonological abstractness. This result would make a strong case for the origins of allophonic variation being in the phonology itself (Fruehwald, 2016; Labov *et al.*, 2016).

III. METHODS

A. Participants

A total of 122 undergraduate students enrolled at Ohio State University completed the experiment. Twenty-one participants who reported that English was not their native language were not included in the analysis. Additionally, seven participants who reported having a history of a speech, hearing, or language disorder were excluded from the analysis.

I examined the remaining 94 participants' self-reported residential history before 18 years old to classify each participant as (1) a resident of only Ohio's Midland region, (2) a resident of only Ohio's Northern region, or (3) neither. Fifty-four participants who did not meet the inclusion criteria for the first two classification categories were excluded from the analysis, leaving 40 participants (22 Midland, 18 Northern). To ensure that the two listener groups were approximately of equal size for the purpose of the analysis, I excluded data from the two Midland participants who participated in the study chronologically last, leaving 20 Midland and 18 Northern participants for the final analysis.

These 38 participants were between 18 and 24 years old ($M = 18.25$ years old), comprising 25 women, 12 men, and one person who declined to report their gender. The ethnoracial breakdown of the participants was as follows: five Asian/Pacific Islander, three biracial, five black/African American, one Hispanic/Latino, and 24 white/Caucasian. The Midland participants grew up in the Cincinnati, Columbus, and Dayton metropolitan areas¹ in Ohio's Midland region, while the Northern participants grew up in the Akron, Canton-Massillon, Cleveland, and Youngstown-Warren metropolitan areas² in Ohio's Northern region. Figure 2 shows this geographic distribution of participants, with darker shading indicating more participants from that location.

B. Materials

The auditory stimuli for this experiment consisted of recordings by two stimulus talkers. Both stimulus talkers were native English speakers who naturalistically produced /ai/ and /æ/ allophony in their dialects. The first talker was a 35 year old woman from Marion County, Ohio (north of the Columbus metropolitan area). This talker is from an area near the North-Midland border in Ohio, which is why her

TABLE I. Representative examples for match, allophonic mismatch, and phonemic mismatch pair types.

Pair type	Definition	Sample pairs
Match	Same phoneme, same phonological context	<i>write-rice</i> <i>ride-rise</i> <i>rat-rack</i> <i>ram-rang</i>
Allophonic mismatch	Same phoneme, different phonological context	<i>write-rise</i> <i>rat-rang</i>
Phonemic mismatch	Different phoneme	<i>write-ram</i> <i>write-rack</i> <i>ride-ram</i> <i>ride-rack</i>

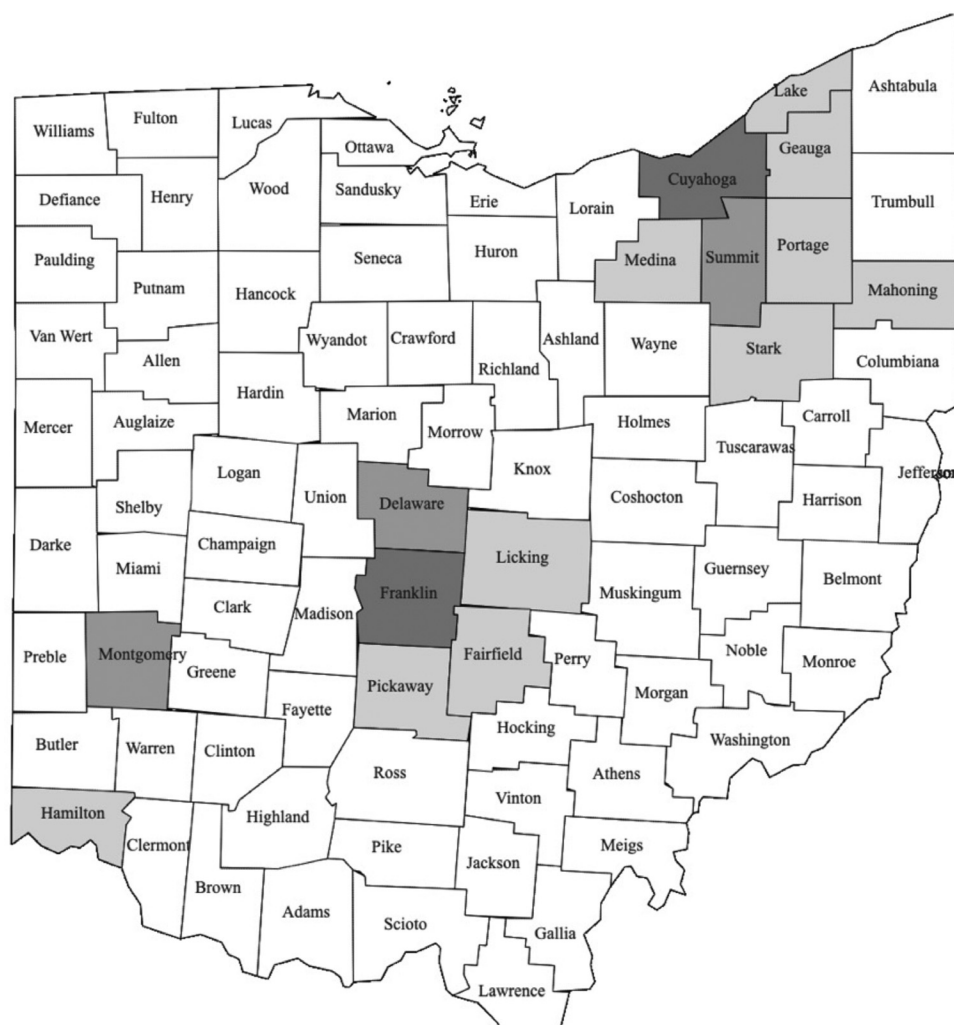


FIG. 2. The geographic distribution of the 38 analyzed participants, with the darkest shade corresponding to five or more participants, the middle shade corresponding to three or four participants, and the lightest shade corresponding to one or two participants.

dialect contains both Northern elements (/ai/ allophony) and Midland elements (/æ/ allophony). The second stimulus talker was a 29 year old woman from Midland County, Michigan (north of the Detroit metropolitan area). This talker is from the Northern dialect region but crucially a part of this region where the Northern Cities Shift (and, therefore, TRAP raising) is receding (Nesbitt *et al.*, 2019; Nesbitt, 2023). Her dialect is characterized by a remaining Northern element (/ai/ allophony) and a Midland element (/æ/ allophony) brought upon by the decline of the Northern Cities Shift in much of Michigan. While these two stimulus talkers are not from the same place, neither stimulus talker's dialect is fully canonically Northern or Midland in nature; instead, both stimulus talkers' dialects combine the same two elements (one Northern and one Midland). Therefore, both stimulus talkers have the same "dialect" for the /ai/ and /æ/ variables that are the focus of this study, though other aspects of their dialects likely vary. Because listeners in this study heard only the stimulus talkers' /ai/ and /æ/ vowels, I treat the two stimulus talkers as having comparable dialects. Figures 3 and 4 show the acoustic characteristics of each

stimulus talker's /ai/ allophony and /æ/ allophony, respectively.

The stimulus talkers' recordings were created in a sound-attenuated booth with a Shure microphone (Shure Inc., Niles, Illinois) placed approximately five inches away from their mouths. Recordings were made using Audacity software (Audacity Team, 2024) with a sampling rate of 44 100 Hz. The stimulus talkers were instructed to read each word aloud in citation form several times in a randomized order. Audio files were processed and manually segmented in Praat (Boersma and Weenink, 2024) following the conventions established by Peterson and Lehiste (1960). All segmented files also underwent root mean square amplitude leveling to 65 dB in Praat.

There were 384 word pairs generated for this experiment, and all participants heard all word pairs exactly once in a randomized order over the course of the experiment. Within each word pair, both words were produced by the same talker. Lexical restrictions caused some word types to be repeated across multiple word pairs, but each repetition was a unique token. Word pairs were manually checked to

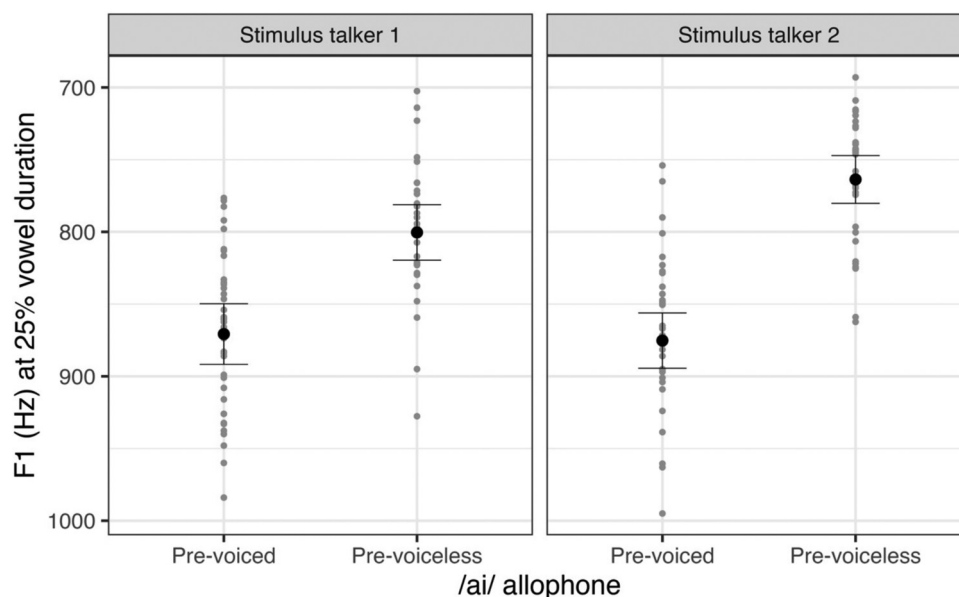


FIG. 3. Acoustic characteristics of pre-voiced and pre-voiceless /ai/ productions for both stimulus talkers, with large black dots showing overall means, smaller gray dots showing individual word means, and error bars showing standard errors of by-word means. Both stimulus talkers clear the Labov *et al.* (2006) 60 Hz threshold for the diagnosis of /ai/ raising (Stimulus talker 1: $M = 871$ Hz pre-voiced and $M = 800$ Hz pre-voiceless F1 at 25% vowel duration; Stimulus talker 2: $M = 875$ Hz pre-voiced and $M = 764$ Hz pre-voiceless F1 at 25% vowel duration).

minimize semantic relatedness (e.g., *bam-bang* was not included because both words are onomatopoeic words that relate to loud noises). Each stimulus talker contributed exactly 192 word pairs to the experimental list, and these word pairs were identical across both stimulus talkers except for a reversed order of presentation (e.g., for the pair *bike* and *bite*, *bike-bite* was the order of presentation for stimulus talker one, and *bite-bike* was the order of presentation for stimulus talker two). Table II shows the characteristics of

the stimuli. Each stimulus talker contributed exactly half of the stimuli for each pair type and subtype. To mitigate potential order effects, for allophonic mismatch and phonemic mismatch pairs, each allophone/phoneme appeared equally often in first versus second position for each stimulus talker.

As Table II suggests, the phonemic mismatch pairs were never minimal pairs. For example, the phonemic mismatch pair *write-rack* appeared in the stimulus list, but the

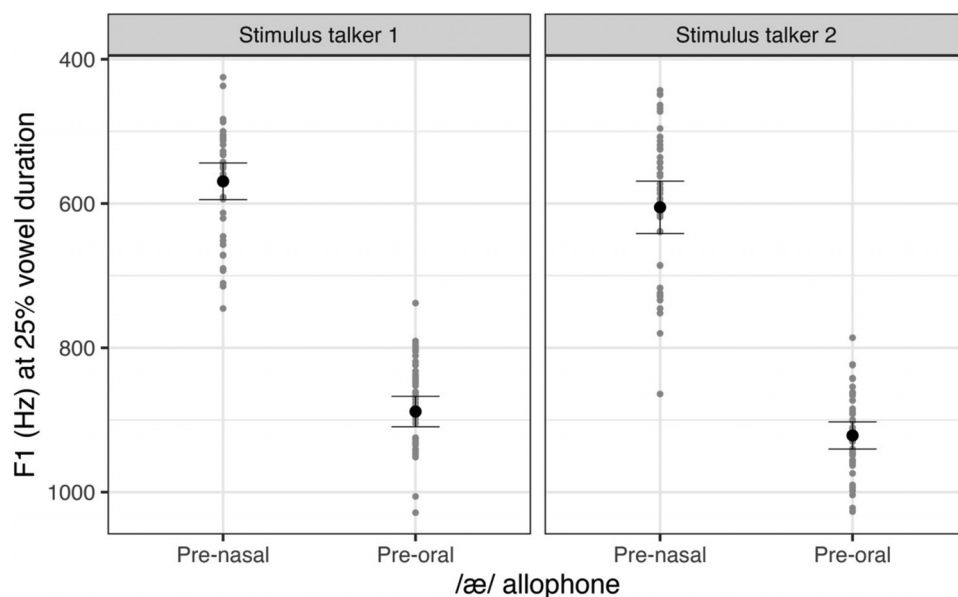


FIG. 4. Acoustic characteristics of pre-voiced and pre-voiceless /æ/ productions for both stimulus talkers, with large black dots showing overall means, smaller gray dots showing individual word means, and error bars showing standard errors of by-word means. Both stimulus talkers show a clear allophonic split by the nasality of the preceding vowel (Stimulus talker 1: $M = 569$ Hz pre-nasal and $M = 888$ Hz pre-oral F1 at 25% vowel duration; Stimulus talker 2: $M = 605$ Hz pre-nasal and $M = 921$ Hz pre-oral F1 at 25% vowel duration).

TABLE II. Stimuli characteristics by pair type and subtype.

Pair type	Number of pairs	Subtypes
Match	128	32 pre-voiceless /ai/ (<i>write-rice</i>) 32 pre-voiced /ai/ (<i>ride-rise</i>) 32 pre-oral /æ/ (<i>rat-rack</i>) 32 pre-nasal /æ/ (<i>ram-rang</i>)
Allophonic mismatch	128	64 /ai/ (<i>write-rise</i>) 64 /æ/ (<i>rat-rang</i>)
Phonemic mismatch	128	32 pre-voiceless /ai/ and pre-nasal /æ/ (<i>write-ram</i>) 32 pre-voiceless /ai/ and pre-oral /æ/ (<i>write-rack</i>) 32 pre-voiced /ai/ and pre-nasal /æ/ (<i>ride-ram</i>) 32 pre-voiced /ai/ and pre-oral /æ/ (<i>ride-rack</i>)

phonemic mismatch pair *write-rat* did not. I restricted the phonemic mismatch pairs in this way because I wanted to preserve as much uniformity across the pair types as possible (except for the vowels, which were being manipulated). Definitionally, the allophonic mismatch pairs were required to have mismatching codas to generate the phonologically conditioned allophonic distinction. The match pairs were also required to have mismatching codas to avoid word repetition within pairs while retaining shared onsets (e.g., *write-rice* instead of *write-write*). While the phonemic mismatch pairs in theory could contain either matching codas (*write-rat*) or mismatching codas (*write-ram*), I only used the latter type of stimulus pairs to ensure that there was no confound between matching/mismatching coda and pair type.³

C. Procedure

Participants sat at individual desktop computers with partitions to maintain privacy during the experiment. First, each participant completed a demographic questionnaire that contained questions about their age, gender, ethnoracial identity, native language(s), residential history, and history of speech, hearing, and language disorders. Then, each participant put on high-quality, over-ear headphones for the rest of the experiment. Participants were instructed both verbally and in writing that they would hear pairs of words aloud over their headphones and that their task was to rate each pair for how similar the two words' vowels sounded using radio buttons that were displayed on the screen. The radio buttons were numbered on a 1–5 scale, with “least similar” displayed under the leftmost radio button (1) and “most similar” displayed under the rightmost radio button (5). These text labels were present throughout the experiment to remind participants about the nature of the radio button scale. Participants registered their rating for each pair by clicking on the corresponding radio button on the screen. However, the software also allowed participants to select intermediate values between whole numbers due to treating the horizontally arranged radio buttons as a continuous

TABLE III. ANOVA table for the /ai/ model.

Coefficient	F-value	Degrees of freedom (numerator, denominator)	p-value
Pair type	4639.12	2, 112.7	<0.001
Listener region	2.45	1, 36.1	0.13
Stimulus talker	1.78	1, 8431	0.18
Pair type × listener region	33.17	2, 8321.5	<0.001

scale. The inter-stimulus interval within word pairs was set to 500 ms and the inter-trial interval between word pairs was set at 1000 ms. Participants took self-timed breaks every 48 trials (approximately every 2 min). The experiment was constructed and run with PsychoPy software (Peirce *et al.*, 2019) on computers running a Windows operating system.

D. Analysis

Though participants were prompted to select whole numbers 1–5 from the on-screen when entering their responses, they were also able to select intermediate values between whole numbers (because doing so was permitted by the experimental software). While ordered scale values would conventionally be treated as being ordinal in nature, the intermediate nature of many of the perceptual similarity ratings raised questions about whether ordinal regression was appropriate. This statistical modelling approach would require rounding participant responses to the nearest whole number for ordinal regression; however, I believe this transformation would reduce meaningful granularity in the data. It is not clear that rounding is equally justifiable across cases (e.g., it seems likely that 3.018519 is intended to be “3,” but it is not clear that 4.762963 is intended to be “5”). I erred on the side of caution by not making these judgments, and therefore I chose not to employ ordinal regression models for this data set. Instead, I adopted the statistical approach of Boomershine *et al.* (2008) to analyzing perceptual similarity ratings data. They z-scored perceptual similarity ratings to help their data more closely meet the assumptions of parametric statistical tests, a strategy that I adopt for my analyses here. This normalization procedure also preserved the granularity of perceptual similarity ratings present in the untransformed data set.

I fit two linear mixed-effects regression models that analyzed variability in perceptual similarity ratings, one per vowel. The models were built and run using R Software (R Core Team, 2024) and the *lme4* package (Bates *et al.*, 2015). Both models were built with the maximal random effects structures that were supported by the complexity of the data, consistent with the conventions established by Barr *et al.* (2013). The data that was analyzed and the list of experimental stimuli are available in full here: <https://osf.io/2gh8a/>.

I fit two models predicting z-scored perceptual similarity ratings using pair type [match, allophonic mismatch (reference level), phonemic mismatch], listener region

TABLE IV. The model output for the /ai/ model. The units of the β -value are not to the original scale due to how the perceptual similarity ratings were z-scored for this analysis.

Coefficient	β -value	Degrees of freedom	t -value	p -value
Intercept	0.72	43.8	11.43	<0.001
Pair type (match)	0.30	207.5	10.57	<0.001
Pair type (phonemic mismatch)	-1.40	208.9	-57.84	<0.001
Region (North)	-0.17	39.4	-1.93	0.06
Stimulus talker (ST2)	-0.01	8431	-1.33	0.18
Pair type (match): region (North)	0.16	8321.2	5.22	<0.001
Pair type (phonemic mismatch): Region (Midland)	-0.06	8321.5	-2.11	0.03

[Midland (reference level), North], and their interaction. Each model analyzed all three pair types, with one model containing all /ai/ data and the other model containing all /æ/ data. All fixed effects were reference level-coded for clarity. Both models contained a covariate of stimulus talker [stimulus talker 1 (reference level), stimulus talker 2] to capture any baseline difference in perceptual similarity ratings that is attributable to acoustic differences between the two stimulus talkers. Both models also contained random by-subject and by-item intercepts.

I tested the significance of fixed effects and their interactions with analysis of variance (ANOVA) tables by running R's `anova()` function on the model. I chose this procedure because the model output alone is not informative for assessing main effects when fixed effects are reference level-coded. In this coding scheme, the model output shows simple effects that represent the effect of a fixed effect only at the reference level of the other fixed effects in the model. The ANOVA procedure bypasses this issue by generating a single F -statistic for each fixed effect, interaction, and covariate. A significant F -statistic represents a significant main effect for a coefficient in the model. All significant main effects were further investigated with *post hoc* tests of estimated marginal means with Tukey

adjustments for multiple comparisons using the *emmeans* package (Lenth, 2020).

IV. RESULTS

A. Analysis of pair type and listener region for match, allophonic mismatch, and phonemic mismatch pairs containing /ai/

The ANOVA table for this model revealed a significant main effect of pair type [$F(2, 112.7) = 4639.12, p < 0.001$] and a significant interaction between pair type and listener region [$F(2, 8321.5) = 33.17, p < 0.001$], as shown in Table III. The main effect of listener region was not significant [$F(1, 36.1) = 2.45, p = 0.13$], suggesting that there was no evidence of a baseline difference in the perceptual similarity ratings provided by listeners across the two regions. The stimulus talker covariate was also not significant [$F(1, 8431) = 1.78, p = 0.18$], providing no evidence that there was a baseline difference in perceptual similarity ratings based on acoustic differences between the two stimulus talkers. The model output itself, with p -values generated by the *lmerTest* package (Kuznetsova *et al.*, 2017), is shown in Table IV.

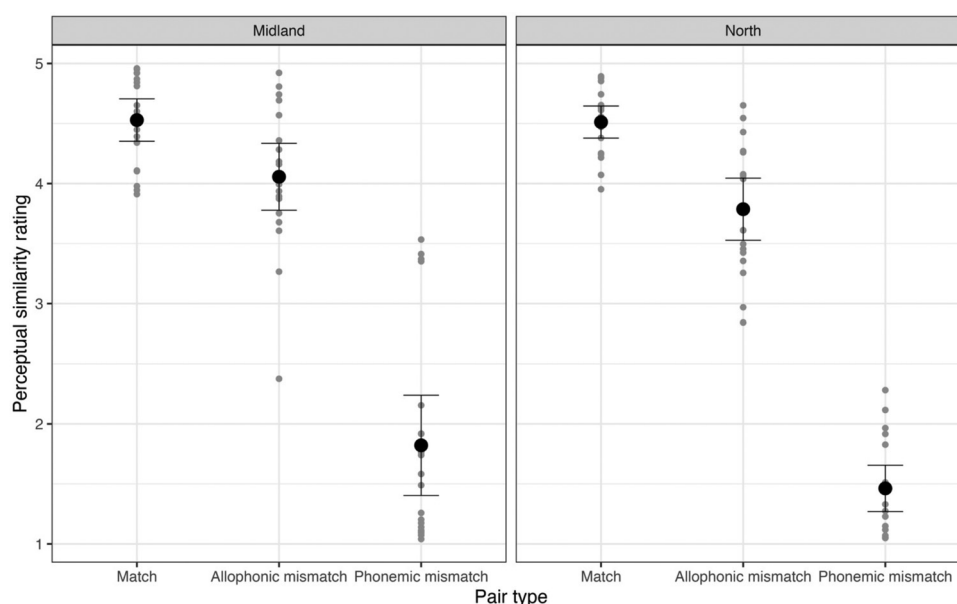


FIG. 5. The interaction between pair type and listener region in the /ai/ model. The error bars represent standard errors of by-subject means, and the gray dots represent individual subject means.

TABLE V. ANOVA table for the /æ/ model.

Coefficient	F-value	Degrees of freedom (numerator, denominator)	p-value
Pair type	775.59	2, 117.4	<0.001
Listener region	0.04	1, 36.1	0.84
Stimulus talker	0.72	1, 8383.5	0.40
Pair type × listener region	155.29	2, 8359	<0.001

To further examine the significant main effect of pair type, I conducted *post hoc* tests of estimated marginal means with Tukey corrections for multiple comparisons. These tests revealed the tripartite structure of perceptual similarity judgments. Listeners perceived match pairs to be more similar than both allophonic mismatch pairs [Midland: $t(207.5) = 10.57$, $p < 0.001$; North: $t(231) = 15.73$, $p < 0.001$] and phonemic mismatch pairs [Midland: $t(210.4) = 69.79$, $p < 0.001$; North: $t(233.9) = 76.09$, $p < 0.001$]. Listeners also perceived allophonic mismatch pairs to be more similar than phonemic mismatch pairs [Midland: $t(208.9) = 57.84$, $p < 0.001$; North: $t(231.9) = 58.19$, $p < 0.001$]. These results demonstrate the positive relationship between perceptual similarity and degree of phonological relatedness, consistent with my prediction and with the observations of Boomershine *et al.* (2008) about perceptual similarity ratings of consonants. Additionally, these results show that all listeners in the study possessed a baseline sensitivity to the contrast among the three pair types regardless of the nature of their exposure.

Post hoc tests of estimated marginal means examining the interaction between pair type and listener region provided additional insights into how listeners' dialect exposure structures their perceptions of the contrasts among the three pair types in this experiment, especially the contrast between the match and allophonic mismatch pairs. These tests specifically probed the magnitude of this match-allophonic mismatch contrast across the two listener region groups. Ultimately, these tests revealed that the magnitude of the difference between perceptual similarity ratings for match and allophonic mismatch pairs was larger for Northern listeners ($\beta = 0.46$, $z = 15.73$, $p < 0.001$) than for Midland listeners ($\beta = 0.30$, $z = 10.57$, $p < 0.001$). This interaction is shown in detail in Fig. 5. This result is consistent with my prediction that exposure facilitates the

development of more phoneme-like cognitive representations of allophones. Increased perceptual differentiation of match and allophonic mismatch pairs for Northern listeners entails that allophonic mismatch pairs move in the direction of phonemic mismatch pairs on the perceptual similarity ratings scale, since phonemic mismatch pairs are the “floor” of this scale. That is, listeners' representations of allophonic variation appear to warp their perceptions of identical auditory stimuli, with listeners with higher exposure showing more perceptually differentiated categories for different allophones.

B. Analysis of pair type and listener region for match, allophonic mismatch, and phonemic mismatch pairs containing /æ/

The ANOVA table for /æ/ revealed a significant main effect of pair type [$F(2, 117.4) = 775.59$, $p < 0.001$] and a significant interaction between pair type and listener region [$F(2, 8359) = 155.29$, $p < 0.001$], as shown in Table V. Neither the main effect of listener region [$F(1, 36.1) = 0.04$, $p = 0.84$] nor the stimulus talker covariate [$F(1, 8383.5) = 0.72$, $p = 0.40$] was significant. These results neatly mirror the results for the /ai/ model. The model output itself, with *p*-values generated using the *lmerTest* package (Kuznetsova *et al.*, 2017), is shown in Table VI.

To further examine the significant main effect of pair type, I conducted *post hoc* tests of estimated marginal means with Tukey corrections for multiple comparisons. The results of these tests also paralleled the results of these tests for the /ai/ model, revealing a clear distinction among the perceptual similarity ratings for the three pair types. Listeners rated match pairs as more similar than allophonic mismatch pairs [Midland: $t(140.7) = 17.30$, $p < 0.001$; North: $t(146.2) = 13.95$, $p < 0.001$] and phonemic mismatch pairs [Midland: $t(139.7) = 33.59$, $p < 0.001$; North: $t(145.2) = 39.87$, $p < 0.001$]. Finally, listeners rated allophonic mismatch pairs as more similar than phonemic mismatch pairs [Midland: $t(143.2) = 13.67$, $p < 0.001$; North: $t(148.8) = 23.85$, $p < 0.001$]. Listeners were sensitive to the three degrees of phonological relatedness present in the three pair types regardless of exposure, and higher perceptual similarity ratings were associated with higher degrees of phonological relatedness.

TABLE VI. The model output for the /æ/ model. The units of the β -value are not to the original scale due to how the perceptual similarity ratings were z-scored for this analysis.

Coefficient	β -value	Degrees of freedom	t-value	p-value
Intercept	0.11	53.7	1.28	0.21
Pair type (match)	0.94	140.7	17.30	<0.001
Pair type (phonemic mismatch)	-0.64	143.2	-13.67	<0.001
Region (North)	0.24	38.4	2.09	0.04
Stimulus talker (ST2)	-0.10	8383.5	-0.85	0.40
Pair type (match): region (North)	-0.17	8358.2	-5.07	<0.001
Pair type (phonemic mismatch): region (Midland)	-0.49	8359.4	-16.71	<0.001

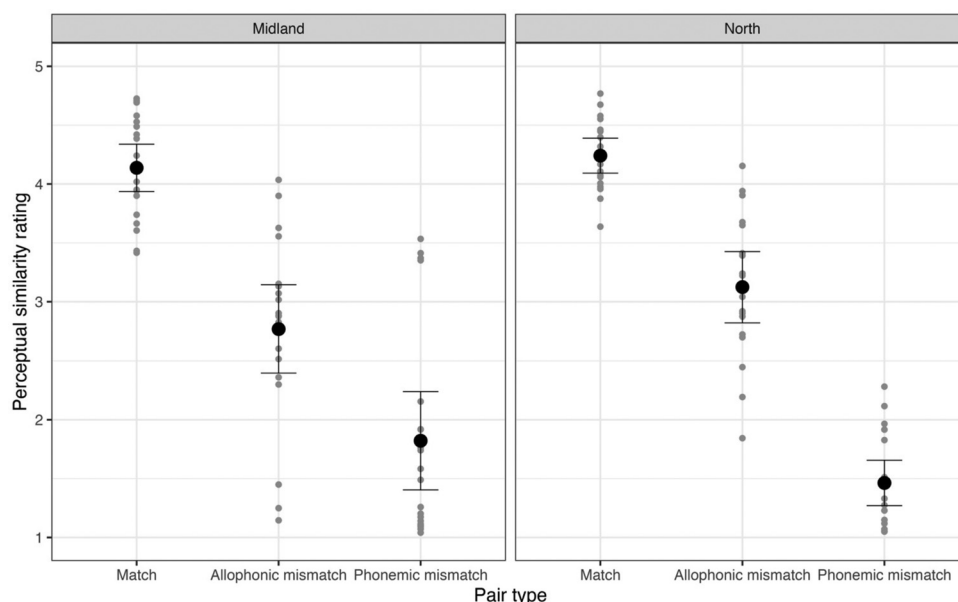


FIG. 6. The interaction between pair type and listener region in the /æ/ model. The error bars represent standard errors of by-subject means, and the gray dots represent individual subject means.

I also conducted *post hoc* tests of estimated marginal means to clarify the nature of the significant interaction between pair type and listener region, especially the nature of differences in the match-allophonic mismatch contrast across the two listener region groups. Specifically, I was interested in investigating whether the magnitude of the difference between these two pair types varied by listener region. These tests revealed that the magnitude of the difference between perceptual similarity ratings for match and allophonic mismatch pairs was larger for Midland listeners ($\beta = 0.94$, $z = 17.30$, $p < 0.001$) than for Northern listeners ($\beta = 0.77$, $z = 13.95$, $p < 0.001$). This interaction is shown in detail in Fig. 6. As was true of the /ai/ model, it appears that higher exposure to dialect-specific allophonic variation is associated with more perceptual differentiation of mismatching allophones. The warping effect that was present in the /ai/ model goes in the opposite direction for the /æ/ model, since for this variable Midland listeners have more exposure to dialect-specific allophonic variation. The consistency and direction of these interaction effects across the two models provide support for the idea that exposure is crucial for the cognitive development of allophonic representations.

V. DISCUSSION

The results from these analyses suggest that listeners possess a broad perceptual sensitivity to allophony across dialect backgrounds, matching the findings of Boomershine *et al.* (2008) about /t d/ allophony. These results are also consistent with evidence that listeners attend to within-phoneme category variation for other reasons, such as for disambiguating lexical items during lexical processing tasks (Beddor *et al.*, 2013; McMurray *et al.*, 2008), for imitation

tasks (Zellou *et al.*, 2016; Zellou *et al.*, 2017), or for ostensible categorical perception tasks (McMurray, 2022).

However, the results also demonstrate that dialect exposure has an additional effect on perceptual evaluations of allophony beyond the baseline amount of perceptual sensitivity to allophonic variation that all listeners show regardless of dialect background. The extent to which listeners perceptually differentiated match pairs versus allophonic mismatch pairs was structured by their exposure to dialect-specific allophonic variation for each vowel. For both the /ai/ and /æ/ vowel, listeners with more exposure to dialect-specific allophonic variation for that vowel rated allophonic mismatch pairs as less similar to match pairs relative to listeners with less exposure to dialect-specific allophonic variation for that vowel. The robustness and uniformity of this pattern across both linguistic variables that were analyzed suggests that synchronic exposure to dialect-specific allophonic variation is likely responsible for generating these perceptual patterns. That is, increased exposure to dialect-specific allophonic variation for a specific linguistic variable is associated with the construction and maintenance of an allophonic representation that is more phoneme-like in nature (and, consequently, less identity-like). The uniformity of these results is especially striking given the acoustic differences present across the two stimulus talkers (see Figs. 3 and 4). Because my analyses revealed no significant effect of stimulus talker on perceptual similarity ratings, they lend support to the idea that listeners' own pre-existing representations of these sounds exerted a relatively uniform, top-down influence on their perceptual similarity ratings that persisted despite the acoustic differences present across the two stimulus talkers.

These results reiterate the results and conclusions of Nesbitt (2023) with a novel task. She reported a strong

perception-production link for /æ/ allophony, arguing that phonological abstractness accumulates over time such that individuals who produced a less advanced version of a sound change in progress also perceived a less advanced version of that sound change when listening to themselves. Conversely, individuals who produced a more advanced version of a sound change also perceived a more advanced version of that sound change when listening to themselves. This synchronization of perception and production led Nesbitt (2023) to interpret her results in a way that supported the theory that phonological abstractness accumulates gradually over the course of a sound change in a community as allophonic productions acoustically diverge more and more. That is, she viewed her data as supporting a late abstractness theory of phonological innovation (Berkson *et al.*, 2017; Davis *et al.*, 2020) rather than early abstractness theory of phonological innovation (Fruehwald, 2016), since the presence of early abstractness would render perceptual similarity ratings equivalent regardless of variability in production. This behavioral equivalence is fundamental to an early abstractness theory, in which changes in production originate in the structures of the phonology itself (e.g., representations of pre-fortis clipping) and should be shared among communities of language users (Fruehwald, 2016; Labov *et al.*, 2016).

My methodology crucially differed from Nesbitt's (2023) methodology by focusing on isolating evidence of divergent representations in perception by removing production variability entirely from the task. Listeners with different types of exposure to dialect-specific allophonic variation were prompted to evaluate identical productions from the same two stimulus talkers who produced both /ai/ and /æ/ allophony. I chose this design to observe perceptual warping effects (Liberman *et al.*, 1967; Pisoni, 1973), in which continuous acoustic space is divided up by listeners due to their existing, top-down representations of speech sound categories. These effects should be observable in a perceptual similarity task, like the one I employed in this study, given that allophonic category boundaries in listeners' representations can influence perceptual comparisons that would otherwise be faithful or veridical to the acoustics of the speech signal. Ultimately, the results of this study demonstrate that listeners with different dialect backgrounds experience perceptual warping of the acoustics of the speech signal due to their varying representations of allophonic categories. Specifically, listeners with lower exposure to dialect-specific allophonic variation for a specific vowel perceived allophonic mismatch pairs as more similar to match pairs than listeners with higher exposure to dialect-specific allophonic variation for that vowel, even when listening to identical stimuli. The most intuitive explanation for this observation is that listeners with different amounts of exposure to dialect-specific allophonic variation for a vowel have different representations of that allophonic variation (i.e., more developed/abstract versus less developed/abstract).

More phonologically abstract representations among listeners with higher exposure to dialect-specific allophonic

variation relative to listeners with lower exposure to dialect-specific allophonic variation is consistent with gradual development of phonological abstractness over the course of a phonological change in progress in a community. Much like Nesbitt's (2023) results, my results support the theory that phonological abstractness is late because perception is a function of higher versus lower exposure. I did not find any evidence supporting the theory that phonological abstractness is early (Fruehwald, 2016), which would require evidence of similar perceptual similarity evaluations regardless of quality of exposure.

Evidence from the perceptual domain builds on existing debates about the nature of phonological abstractness and its development for individuals. While production data are informative in some cases, conclusions drawn from this data are necessarily limited because they rely on conflicting theoretical ideas of what makes a production pattern "phonological." For example, it is unclear that conflicting theoretical ideas about what makes /ai/ raising phonological (i.e., conventionally, whether [ai] or [aɪ] appears in *writer*) are resolvable with production data alone. Additionally, it is not clear that broad statements about the nature of individuals' phonologies that hinge on this sort of evidence are justifiable given that pre-flap /ai/ productions are so exceedingly rare in English [though, see Hall (2012) for an interesting discussion of variant predictability]. In contrast, perception data provide a broader snapshot of what listeners' representations of allophones are like. Perceptual similarity ratings are associated with degree of phonological contrast (Boomershine *et al.*, 2008). Therefore, it is reasonable to assert that the magnitude of the difference in perceptual similarity ratings between match pairs and allophonic mismatch pairs reflects the magnitude of the representational difference that is present for listeners, though there is evidence that other factors like predictability of distribution (Hall *et al.*, 2015) and context (Murphy, 2019) may also play a substantial role.

While these factors are not the explicit focus of the current study, it is worth considering how they may have impacted the two variables I investigated. I want to especially address the issue of context. Listeners never heard /ai/ allophones in a context in which they potentially formed a minimal pair (i.e., pre-flap), but they did hear many /æ/ allophones in a context in which they potentially formed a minimal pair in this study. Previous research suggests that embedding allophones in their phonological contexts can reduce listeners' perceptual sensitivity to that distinction (Murphy, 2019; Peperkamp *et al.*, 2003), but it is clear from the results of this study that this sort of embedding did not have this effect. If anything, mismatching allophones appear to be more perceptually differentiable than matching allophones for /æ/ than for /ai/ (see Figs. 3 and 4), even though the /æ/ allophones were in a potentially minimal pair-generating phonological context while the /ai/ allophones were not. Despite this discrepancy, the effect of dialect exposure persists in my results, showing a robust influence on perception across both vowel variables.

In the 1–5 perceptual similarity continuum employed in this study, there is a fair amount of representational space available for allophonic mismatch pairs to occupy between match pairs and phonemic mismatch pairs. A listener's exposure to dialect-specific allophonic variation has the effect of widening the gap between match pairs and allophonic mismatch pairs and, therefore, also narrowing the gap between allophonic mismatch pairs and phonemic mismatch pairs. Therefore, given these results, exposure to dialect-specific allophonic variation functions as a pathway for developing representations of allophonic categories that are more phoneme-like in nature, consistent with accumulation of gradient degrees of phonological contrast over time with exposure. Exposure is crucial for allophonic category development, as was proposed by Sumner and Samuel (2009). While all listeners regardless of dialect background differentiated match pairs from allophonic mismatch pairs, the perceptual distance between allophonic categories in listeners' representations is a function of their amount of exposure to dialect-specific allophonic variation. This effect is present despite the different diachronic origins and synchronic profiles of the two cases of allophony that I studied, suggesting that the uniting factor of dialect exposure motivates the construction and maintenance of listeners' cognitive representations of phonological categories.

It is also worth asking whether these two cases of allophony are truly allophonic, or whether one or both have obtained some sort of marginal phoneme status (e.g., Hualde *et al.*, 2017). If this statement were true, then it would potentially be unsurprising to find that listeners perceive so-called robust phonemic distinctions (e.g., *write-rack*) as less similar than marginal phonemic distinctions (e.g., *write-rise*). However, a stronger version of this theoretical position would predict that the existence of a single minimal pair is enough to render the marginal phonemic distinction perceptually equivalent to the so-called robust phonemic distinction. While the current study was not designed to investigate listeners' perceptions of true minimal pairs, a future study that employs true minimal pairs for phonemic mismatch pairs (e.g., *write-rat*) to understand whether they are treated as perceptually equivalent to allophonic mismatch pairs (e.g., *write-rise*) would be worthwhile. Although I was unable to locate evidence that true minimal pairs were evaluated differently than non-minimal pairs in parts of the existing data set (see footnotes), whether this distinction matters for the perceptual evaluation of phonemic mismatch pairs is an empirical question worthy of future study that seems likely to yield additional theoretical clarification about early versus late abstractness and also the relationship between minimal pair-hood and perceptions of phonological contrast.

VI. CONCLUSION

Regardless of their dialect exposure, listeners attend to allophonic variation in the speech signal when evaluating perceptual similarity (Boomershine *et al.*, 2008). While it is

intuitive that allophonic mismatch pairs would be perceived somewhere between match pairs and phonemic mismatch pairs in terms of perceptual similarity in this study (Trubetzkoy, 1939; Boomershine *et al.*, 2008), the extent to which listeners' perceptions would be structured according to their exposure to dialect-specific allophonic variation was key for understanding the development of phonological abstractness over the course of a phonological change in a community. My results are consistent with a late abstractness theory of phonologization due to the existence of representational differences among listeners with different amounts of exposure to dialect-specific allophonic variation. Because these results are also consistent across both variables I studied, it is likely that the shared component of more versus less exposure to dialect-specific allophonic variation is responsible for this uniformity.

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AUTHOR DECLARATIONS

Conflict of Interest

The author has no conflicts to disclose.

Ethics Approval

This research was approved by Ohio State University's Institutional Review Board (No. 2022E0184, No. 2022E0804, and No. 2023B0314).

DATA AVAILABILITY

The data that support the findings of this study are openly available in the Open Science Framework repository at https://osf.io/2gh8a/?view_only=52af7a1274dc4677a9a362353f0c31ba.

¹Reported counties of residence include Delaware, Fairfield, Franklin, Hamilton, Licking, Montgomery, and Pickaway.

²Reported counties of residence include Cuyahoga, Geauga, Lake, Mahoning, Medina, Portage, Stark, and Summit.

³It is possible that using non-minimal pairs (e.g., *write-rack*) for the phonemic mismatch pairs instead of minimal pairs (e.g., *write-rat*) could have artificially reduced the perceptual similarity ratings for phonemic mismatch pairs (by reducing shared phonological content). However, an investigation of the allophonic mismatch pairs that were non-minimal pairs versus minimal pairs revealed no significant difference in perceptual similarity ratings across the two groups ($t = -1.62$, $p = 0.11$). If anything, the difference between the two groups trended in the opposite direction, with true minimal pairs ($M = 2.88$) being evaluated as less perceptually similar than non-minimal pairs ($M = 2.98$).

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